# Mathematical Statistics




2 September 2021

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# Chapter 1

# Gamma Function

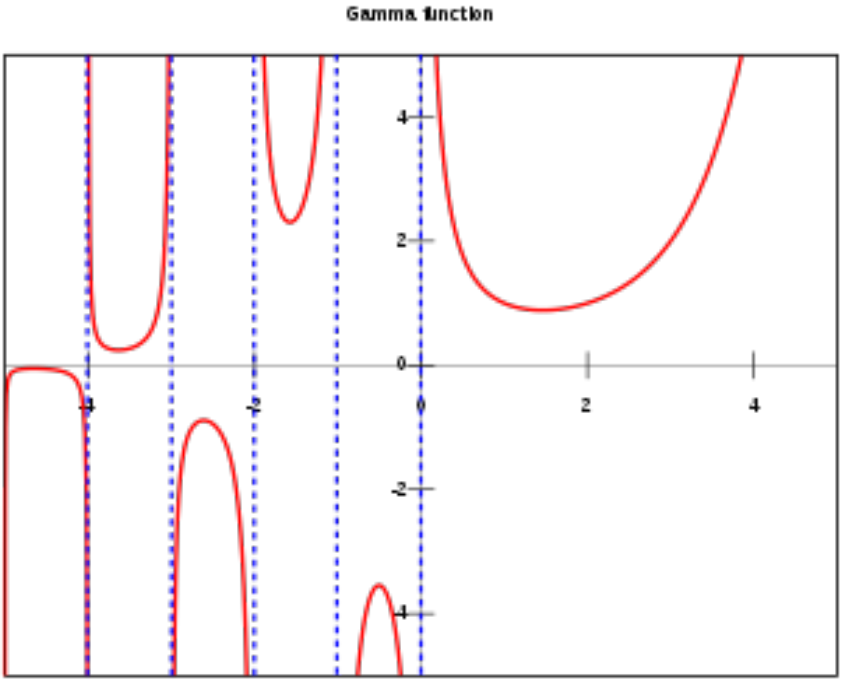
The gamma function is considered an extension of a factorial function

$$n! = n \text{ x } (n-1) \text{ x } \ldots \text{ x } 2 \text{ x } 1$$
$$\Gamma(n) = (n-1)!$$

Derived by Daniel Bernoulli, for complex numbers with a positive real part, the gamma function is defined via a convergent improper integral:

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx \quad , \mathcal{R}(z) > 0 \tag{1.1}$$

The gamma function then is defined as the analytic continuation of this integral function to a meromorphic function that is holomorphic in the whole complex plane except zero and the negative integers, where the function has simple poles.



## 1.1 Proof By Induction

The notation $\Gamma(z)$ is due to Legendre [1] If the real part of the complex number $z$ is strictly positive ( $\Re(z) > 0$ ), then the integral

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$$

converges absolutely, and is known as the Euler integral of the second kind. (Euler's integral of the first kind is the beta function. [1] ) Using integration by parts, one sees that:

$$\Gamma(z+1) = \int_0^\infty x^z e^{-x} dx$$

Use Integration By Parts $\int udv = uv - \int vdu$

Let $u = x^z, du = zx^{z-1}dx$ $dv = e^{-x}dx, v = -e^{-x}$

$$\begin{aligned}
\int_0^\infty &= uv - \int_0^\infty vdu = \underbrace{[-x^z e^{-x}]_0^\infty}_{=0} + \int_0^\infty zx^{z-1}e^{-x}dx \\
&= \left[-x^z e^{-x}\right]_0^\infty + \int_0^\infty zx^{z-1}e^{-x}dx \\
&= \lim_{x\to\infty}\left(-x^z e^{-x}\right) - \left(-0^z e^{-0}\right) + z\int_0^\infty x^{z-1}e^{-x}dx \\
&= z\int_0^\infty x^{z-1}e^{-x}dx
\end{aligned}$$

Recognizing that $-x^z e^{-x} \to 0$ as $x \to \infty$

$$\begin{aligned}
\Gamma(z+1) &= z\int_0^\infty x^{z-1}e^{-x}dx \\
&= z\Gamma(z)
\end{aligned}$$

We can calculate $\Gamma(1)$

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty x^{1-1}e^{-x}dx \\
&= \left[-e^{-x}\right]_0^\infty \\
&= \lim_{x\to\infty}\left(-e^{-x}\right) - \left(-e^{-0}\right) \\
&= 0 - (-1) \\
&= 1
\end{aligned}$$

We can calculate $\Gamma(n)$ for n values

$$\begin{aligned}
\Gamma(2) &= \Gamma(1+1) = 1\Gamma(1) = 1 \\
\Gamma(3) &= \Gamma(2+1) = 2\Gamma(2) = 2 \\
\Gamma(4) &= \Gamma(3+1) = 3\Gamma(3) = 6 \\
\Gamma(5) &= \Gamma(4+1) = 4\Gamma(4) = 24 \\
\Gamma(n) &= (n-1)!
\end{aligned}$$

Perhaps the best-known value of the gamma function at a non-integer argument is

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Lets consider the **Normal Distribution**

$$\begin{aligned}
f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\
&\text{Let's choose } \mu = 0, \sigma^2 = \frac{1}{2} \\
f(x) &= \frac{1}{\sqrt{\pi}} e^{-x^2}
\end{aligned}$$

Lets again consider $\Gamma(\frac{1}{2})$

$$\begin{aligned}
\Gamma(\frac{1}{2}) &= \int_0^\infty x^{1/2} e^{-x} dx \\
u &= \sqrt{x} = x^{1/2} \implies u^2 = x \\
du &= \frac{1}{2} x^{-1/2} dx = \frac{1}{2\sqrt{x}} dx \implies dx = 2\sqrt{x} du \\
\Gamma(\frac{1}{2}) &= \int_0^\infty \frac{1}{\sqrt{x}} e^{-u^2} 2\sqrt{x} \cdot du \\
&= 2 \int_0^\infty e^{-u^2} \cdot du \\
&= 2 \cdot \frac{\sqrt{\pi}}{2} \\
&= \sqrt{\pi}
\end{aligned}$$